APARNA SINGH

Software Developer | AI / ML Engineer | Data Science & Analytics

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SUMMARY

Software Developer and AI/ML Engineer with expertise in Python, Java, JavaScript, machine learning, deep learning, and cloud computing. B.Tech Information Technology student (CGPA 9.03) experienced in end-to-end ML pipelines, REST API development, LLM integration, and scalable web application development. Google GenAI Exchange Semi-Finalist; Top 10 at RIFT '26 Hackathon. Proficient in TensorFlow, Scikit-learn, Keras, OpenCV, NLP, Computer Vision, Docker, CI/CD, and Google Cloud Platform (GCP).

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript (ES6+), SQL, HTML5, CSS3, Bash

AI / ML & Deep Learning: TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face, LangChain, RAG, NLP, Computer Vision, LLM Integration, Classification, Clustering, Regression, Model Evaluation (Precision, Recall, F1, ROC-AUC)

Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Dimensionality Reduction (PCA), Power BI, Excel

Cloud & DevOps: Google Cloud Platform (GCP), Vertex AI, BigQuery, Cloud Run, Docker, Git, GitHub Actions, CI/CD Pipeline

Web & Backend Development: Next.js, Node.js, REST APIs, Firebase, Gemini API, Google Maps API, Google Translation API

Databases: MySQL, Firebase Firestore, BigQuery, SQL (DBMS)

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

EDUCATION

Bharati Vidyapeeth College of Engineering, Pune — July 2023 – Present

Bachelor of Technology, Information Technology | CGPA: 9.03 / 10.0

Tender Hearts School, Lucknow — February 2022

Higher Secondary Certificate (Class XII)

EXPERIENCE

AI/ML Developer — Google GenAI Exchange Hackathon 2025 (Semi-Finalist) — 2025

Tech Stack: Next.js, Node.js, Python, Firebase, Gemini API, Vertex AI, BigQuery, RAG, Google Cloud Platform

- **Developed** KarigarSetu, an AI-driven full-stack cultural marketplace using Next.js, Node.js, and Gemini API with Retrieval-Augmented Generation (RAG) and LRU caching for low-latency conversational search, reducing average query response time by 40%.
- **Architected** scalable GCP backend using Vertex AI for ML model inference and BigQuery for large-scale analytics pipelines; integrated Google Translation and Maps APIs supporting 5+ regional languages and location-aware discovery for artisans across India.

Software Developer — Smart India Hackathon (SIH) 2025 | RIFT '26 Hackathon — Top 10 — 2025 – 2026

National-level hackathons: agriculture domain (SIH) | health-tech application (RIFT '26, 24-hour open innovation sprint)

- **Designed** system architecture and delivered a working prototype for an agriculture-sector problem statement at Smart India Hackathon (SIH) 2025, a national-level government software engineering competition with 50,000+ registered teams.
- **Delivered** a health-tracking web application at RIFT '26 Hackathon (PW IOI), placing in the Top 10 out of 50+ teams in a 24-hour open innovation sprint.

PROJECTS

CreditWise — Loan Approval Prediction System — Python | Machine Learning | GitHub

Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook — Supervised Learning

- **Constructed** an end-to-end supervised ML pipeline for binary loan-approval classification using 3 algorithms — KNN, Logistic Regression, and Naïve Bayes — with full EDA, feature engineering, and data preprocessing on real-world financial data.
- **Achieved** 92% classification accuracy (Logistic Regression); evaluated all 3 models via Precision, Recall, F1-Score, and ROC-AUC, producing results suitable for automated financial decision-support systems.

SmartCart — Customer Segmentation System — Python | Machine Learning | GitHub

Scikit-learn, Pandas, Matplotlib, Seaborn, Jupyter Notebook — Unsupervised Learning, K-Means, PCA

- **Engineered** an intelligent customer segmentation system using K-Means clustering and PCA (dimensionality reduction) to uncover customer behaviour patterns within a 2,000+ record dataset, enabling targeted customer engagement and higher retention rates across 4+ identified customer segments.

CERTIFICATIONS

- Cloud Computing — IIT Madras, NPTEL / Swayam (November 2024)
- Database Management Systems (DBMS) — IIT Kharagpur, NPTEL / Swayam (March 2025)
- GenAI Powered Data Analytics Job Simulation — TATA Forage (November 2025)

ACTIVITIES & LEADERSHIP

- Technical Member, GeeksforGeeks Campus Body, Bharati Vidyapeeth — organized coding contests, DSA workshops, and tech talks (April 2025 – Present).
- UNESCO Innovation Program 2025 — Contributed to the Moody Project, supporting problem definition, solution design, and social-impact planning.